cl_px4_mr::CbLawnmower

+ CbLawnmower()
+ CbLawnmower()
+ setParams()
+ params()
+ setOrigin()
# buildPath()
# behaviorName()

cl_px4_mr::CbPx4PathFollowerBase

+ CbPx4PathFollowerBase()
+ ~CbPx4PathFollowerBase()
+ onStateOrthogonalAllocation()
+ setFollowerParams()
+ followerParams()
+ onEntry()
+ onExit()
+ update()
# buildPath()
# behaviorName()
# onPathStarted()
# onPathCompleted()
# totalLength()
# progressFraction()
# carrotVertexIndex()
- commandFor()
- applyCentering()
- tangentYawAt()

cl_px4_mr::CbPx4ClientBehaviorBase

+ onStateOrthogonalAllocation()
+ ~CbPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()

smacc2::SmaccAsyncClient
Behavior

+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()

smacc2::ISmaccClientBehavior

+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()

smacc2::ISmaccStateMachine

+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration
Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()

-stateMachine_

smacc2::SignalDetector

+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()

-smaccStateMachine_

smacc2::ISmaccOrthogonal

+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()

-stateMachine_

smacc2::ISmaccComponent

+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()

#stateMachine_

smacc2::ISmaccClient

+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()

-stateMachine_

SharedPtr

#nh_

smacc2::ISmaccState

+ ~ISmaccState()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition
TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()

#node_

SharedPtr

#timer_

SharedPtr

#eventsLogPub_
#stateMachinePub

_

#stateMachineStatus
Pub_
#transitionHistory
Service_
#transitionLogPub_

-statusPub_

cl_px4_mr::CpVehicleCommand

+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()

-publisher_

cl_px4_mr::CpVehicleStatus

+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()

-subscriber_

cl_px4_mr::CpTrajectorySetpoint

+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()

-publisher_

cl_px4_mr::CpVehicleLocal
Position

+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()

-subscriber_

cl_px4_mr::CpOffboardKeep
Alive

+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()

-publisher_

cl_px4_mr::CpTunnelCentering

+ CpTunnelCentering()
+ ~CpTunnelCentering()
+ onInitialize()
+ lateralOffsetM()
+ verticalOffsetM()
+ valid()
+ leftWallM()
+ rightWallM()
+ floorM()
+ ceilingM()
+ params()
- onCloud()
- onAttitude()

-attitudeSub_

smacc2::client_core
_components::CpTopicSubscriber
< sensor_msgs::msg::PointCloud2 >

+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()

-sub_

vector< smacc2::ISmaccState * >

#currentState_

shared_ptr< smacc2
::introspection::SmaccState
Info >

#currentStateInfo_

SmaccStatus

#status_msg_

map< std::string, std
::shared_ptr< smacc2
::ISmaccOrthogonal > >

#orthogonals_

shared_ptr< smacc2
::introspection::SmaccState
MachineInfo >

#stateMachineInfo_

recursive_mutex

-eventQueueMutex_
-m_mutex_

atomic< StateMachineInternal
Action >

-stateMachineCurrentAction

map< void *, std::shared
_ptr< smacc2::CallbackCounterSemaphore > >

-stateCallbackConnections

map< std::string, std
::pair< std::function
< std::string()>, boost
::any > >

-globalData_

vector< smacc2_msgs
::msg::SmaccTransitionLogEntry >

-transitionLogHistory_

SMRunMode

-runMode_

-signalDetector_

vector< smacc2::ISmaccUpdatable * >

-updatableClients_

vector< std::vector
< smacc2::ISmaccUpdatable * > >

-updatableStateElements_

atomic< int64_t >

-timeoutMs_

-lastCloudNs_ -lastState_

double

-loop_rate_hz

-refLat_
-refLon_

cl_px4_mr::MapProjection

+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()

-refCosLat_
-refLat_
-refLon_
-refSinLat_

atomic< bool >

-active_

-completed_

-isShutdownRequested_

-end_
-initialized_
-rosInitialized_

SmaccFifoScheduler *

-scheduler_

processor_handle

-processorHandle_

thread

-signalDetectorThread_

ExecutionModel

-executionModel_

int64_t

-stateSeqCounter_

-currentState

#parentState_

shared_ptr< rclcpp
::Logger >

#logger_

vector< std::shared
_ptr< smacc2::StateReactor > >

#stateReactors_

vector< std::shared
_ptr< smacc2::SmaccEventGenerator > >

#eventGenerators_

smacc2::introspection
::SmaccStateInfo

+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()

#stateInfo_

std::enable_shared
_from_this< SmaccStateInfo >

map< const std::type
_info *, std::vector
< smacc2::introspection
::ClientBehaviorInfoEntry > >

+staticBehaviorInfo

map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccStateReactor
Info > > >

+stateReactorsInfo

map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccEventGenerator
Info > > >

+eventGeneratorsInfo

int

-lastProgressDecile_

+depth_
+stateIndex_

cl_px4_mr::TunnelCentering
Params
+ minRangeM
+ wallBandHalfHeightM
+ verticalSearchM
+ wallFadeM
+ smoothing

+minFloorPoints
+minWallPoints

string

+demangledStateName
+fullStateName

-topicName_

smacc2::client_core
_components::CpTopicSubscriber
< MessageType >

+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()

-topicName_

weak_ptr< smacc2::introspection
::SmaccStateMachineInfo >

+stateMachine_

weak_ptr< smacc2::introspection
::SmaccStateInfo >

+parentState_

vector< smacc2::introspection
::SmaccTransitionInfo >

+transitions_

vector< std::shared
_ptr< smacc2::introspection
::SmaccStateInfo > >

+children_

type_info *

+tid_

-currentOrthogonal

-orthogonal_

vector< std::shared
_ptr< smacc2::ISmaccClient > >

#clients_

vector< std::vector
< std::shared_ptr< smacc2
::ISmaccClientBehavior > > >

-clientBehaviors_

mutex

-mutex_

-mutex_

-mutex_

-mutex_

-attitudeMutex_

optional< std::future
< int > >

-onEntryThread_
-onExitThread_

function< void()>

-postFailureEventFn
_

-postFinishEventFn
_

-postSuccessEventFn_

smacc2::SmaccSignal
< void()>

-onFailure_
-onFinished_
-onSuccess_

+onArmed_
+onDisarmed_
+onLanded_
+onModeChanged_

+onLocalPositionReset

_

+onPositionReceived_

cl_px4_mr::CpGoalChecker

+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()

+onGoalReached_

boost::signals2::signal
< void(), optional_last
_value< typename boost::
function_traits< void()>
::result_type >, int, std
::less< int >, function< void
()>, typename extended_signature
< function_traits< void()>::arity,
void()>::function_type, boost::signals2::mutex >

smacc2::SmaccSignal
< Signature, Combiner,
Group, GroupCompare,
SlotFunction, ExtendedSlotFunction,
Mutex >

< void()>

smacc2::SmaccSignal
< void(const sensor
_msgs::msg::PointCloud2 &)>

< void(const sensor
_msgs::msg::PointCloud2 &)>

smacc2::SmaccSignal
< void(const MessageType &)>

< void(const MessageType &)>

boost::signals2::signal
< Signature, optional
_last_value< typename boost
::function_traits< Signature
>::result_type >, int, std
::less< int >, function< Signature
>, typename extended_signature<
function_traits< Signature >::arity,
Signature >::function_type, boost::
signals2::mutex >

smacc2::ISmaccUpdatable

+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()

optional< rclcpp::Duration >

-periodDuration_

optional< rclcpp::Time >

-lastUpdate_

#vehicleCommand_

#owner_

map< smacc2::ComponentKey,
std::shared_ptr< smacc2
::ISmaccComponent > >

#components_

#vehicleStatus_

uint8_t

-armingState_
-navState_
-prevArmingState
_

-prevNavState_

-xyResetCounter_
-zResetCounter_

#trajectorySetpoint_

-trajectorySetpoint_ #localPosition_

-localPosition_ -localPosition_

float

-entryHeading_
-sCarrot_
-totalLen_

-heading_
-refAlt_
-x_
-y_
-z_

-goalX_
-goalY_
-goalZ_
-xyTolerance
_

-zTolerance_

cl_px4_mr::PathFollowerParams

+arrivalXyTol
+arrivalZTol
+autoTimeoutFactor
+fixedYaw
+groundSpeed
+leash
+minSegmentLength

+ceilingClearanceM
+columnHalfWidthM
+floorClearanceM
+floorLookBackM
+lateralSearchM
+lookAheadMaxM
+lookAheadMinM
+maxLateralM
+maxVerticalM
+minFloorClearanceM

...

cl_px4_mr::NedPoint

+x

+y
+yaw
+z

cl_px4_mr::FlightPattern
LawnmowerParams

+altitudeAgl
+laneHeading
+laneLength
+laneSpacing
+originX
+originY
+width

bool

-hasPublished_

-firstMessage_
-headingGoodForControl
_

-valid_
-velocityValid
_

-xyGlobal_
-zGlobal_

-initialized_

-goalActive_

-enabled_

+prependCurrentPosition
+useTunnelCentering

-haveAttitude_

+levelFrame

-firstMessage_
-initialized_

-firstMessage_
-initialized_ +originIsCenter

uint64_t

-refTimestamp_

-refTimestamp_

-projection_

TrajectorySetpoint

-lastSetpoint_

#goalChecker_

#offboardKeepAlive_

optional< std::chrono
::steady_clock::time
_point >

-watchdogStart_

#followerParams_

YawMode

+yawMode

#tunnelCentering_

-params_

-subscriber_

optional< int >

+queueSize

+queueSize

function< void(const
sensor_msgs::msg::PointCloud2 &)>

+postInitialMessageEvent
+postMessageEvent

+onFirstMessageReceived
_

+onMessageReceived_

boost::signals2::signal
< void(const sensor_msgs
::msg::PointCloud2 &), optional
_last_value< typename boost::
function_traits< void(const sensor
_msgs::msg::PointCloud2 &)>::result
_type >, int, std::less< int >, function
< void(const sensor_msgs::msg::PointCloud2
&)>, typename extended_signature< function
_traits< void(const sensor_msgs::msg::PointCloud2
&)>::arity, void(const sensor_msgs::msg::PointCloud2
&)>::function_type, boost::signals2::mutex >

< sensor_msgs::msg
::PointCloud2 >

function< void(const
MessageType &)>

+postInitialMessageEvent
+postMessageEvent

+onFirstMessageReceived
_

+onMessageReceived_

boost::signals2::signal
< void(const MessageType
&), optional_last_value
< typename boost::function
_traits< void(const MessageType
&)>::result_type >, int, std
::less< int >, function< void(const
MessageType &)>, typename extended
_signature< function_traits< void(const
MessageType &)>::arity, void(const MessageType
&)>::function_type, boost::signals2::mutex >

SharedPtr

-sub_

array< float, 4 >

-attitudeQ_

atomic< float >

-ceiling_
-floor_
-lateral_
-left_
-right_
-vertical_

time_point

-lastUpdateTime_

-lastLog_

vector< cl_px4_mr::
NedPoint >

-path_

vector< float >

-cumLen_

atomic< size_t >

-carrotVertex_

-lastCmd_

-params_

Turn

+firstTurn